

OpenFOAM Developer Training, No Fluff!

Ali Shayegh

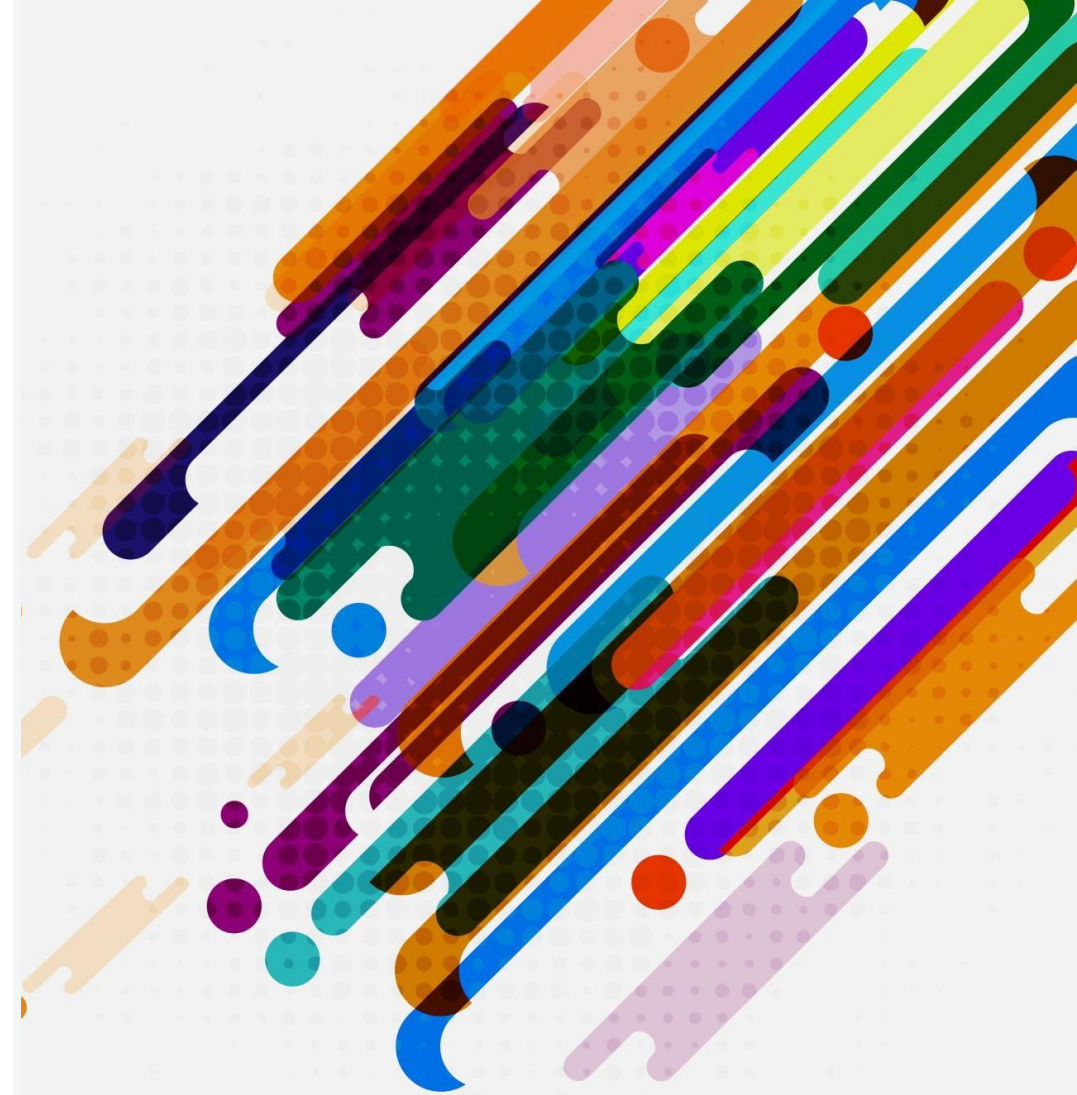
TensorFields, University College Dublin



University
College
Dublin



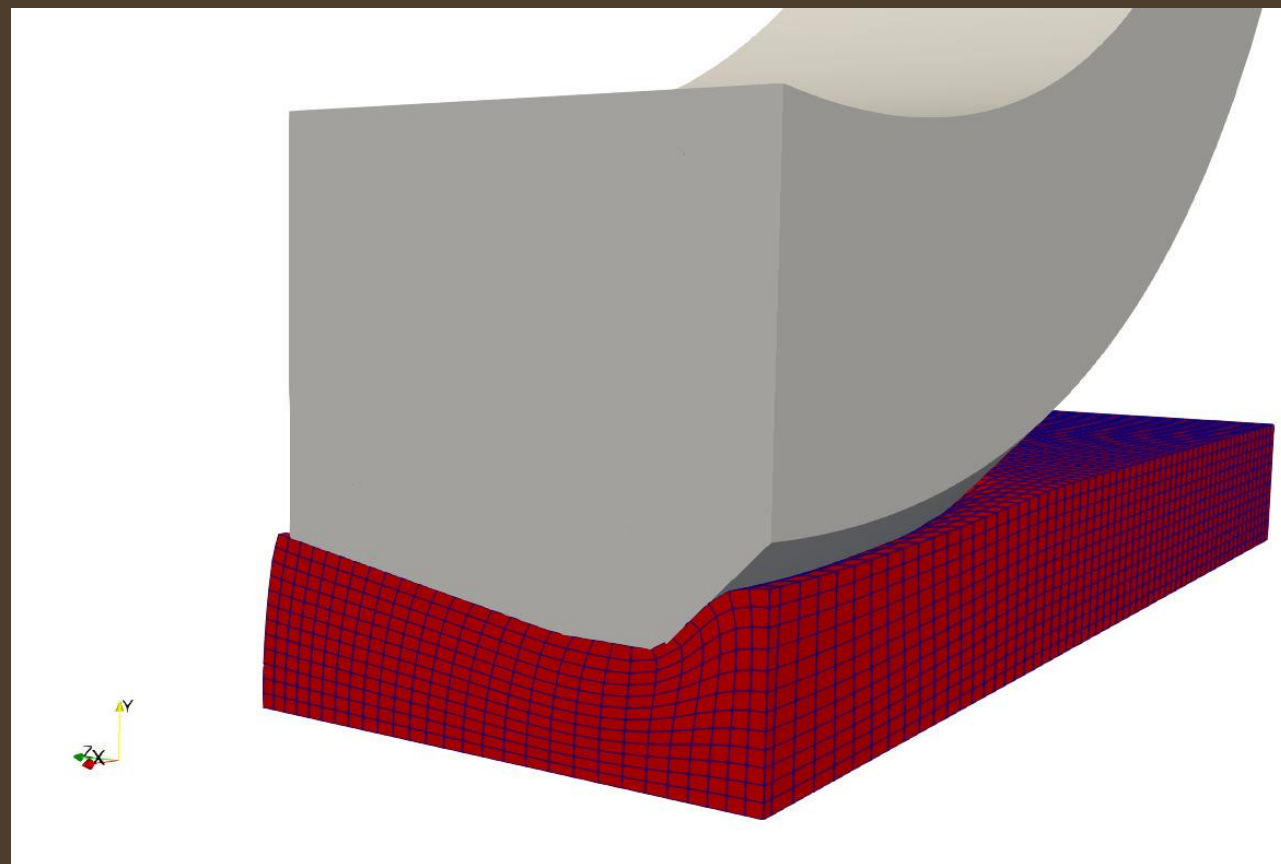
Tensorfields



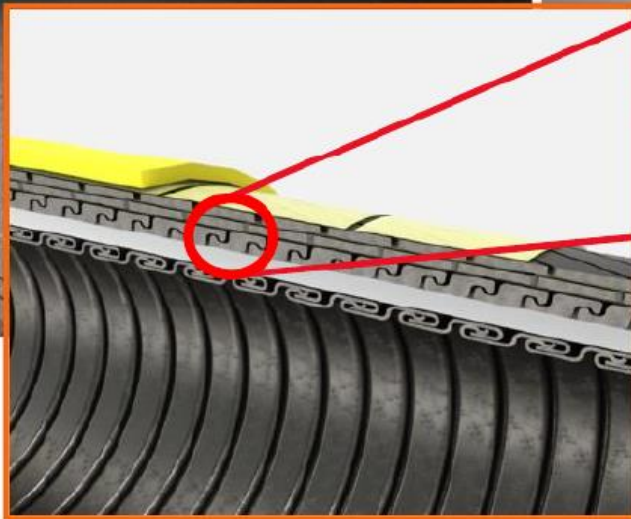
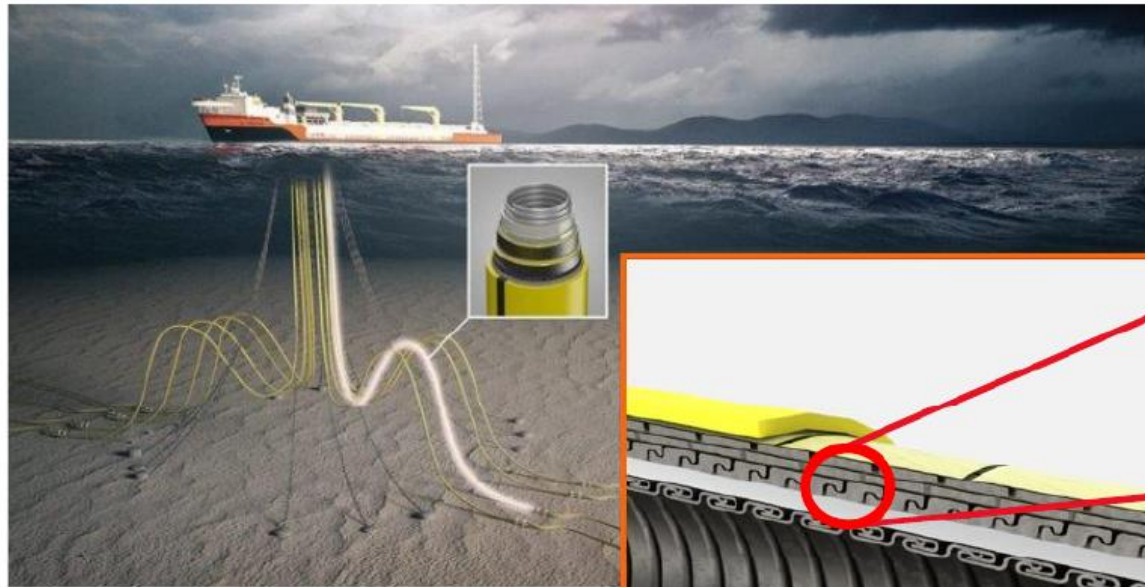
ABOUT US

Ali Shayegh

- CTO at **TensorFields**
- Researcher at **UCD**
- 8+ years of **OpenFOAM** experience
- Senior member on **cfD-online**



ABOUT US



ABOUT US

TensorFields

- Est. 2023
- OpenFOAM **training** initiative
- Physics-based modeling **consultant**
- Based in Dublin, Ireland.
- [**tensorfields.com**](https://tensorfields.com)



Goal

- Create engineering-standard applications.



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Agenda

- Write an application and a library in C++



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- Write an application and a library in C++
- Write a plain bullet point plan for solving a **PDE**



Goal

- Create engineering-standard applications.

Agenda

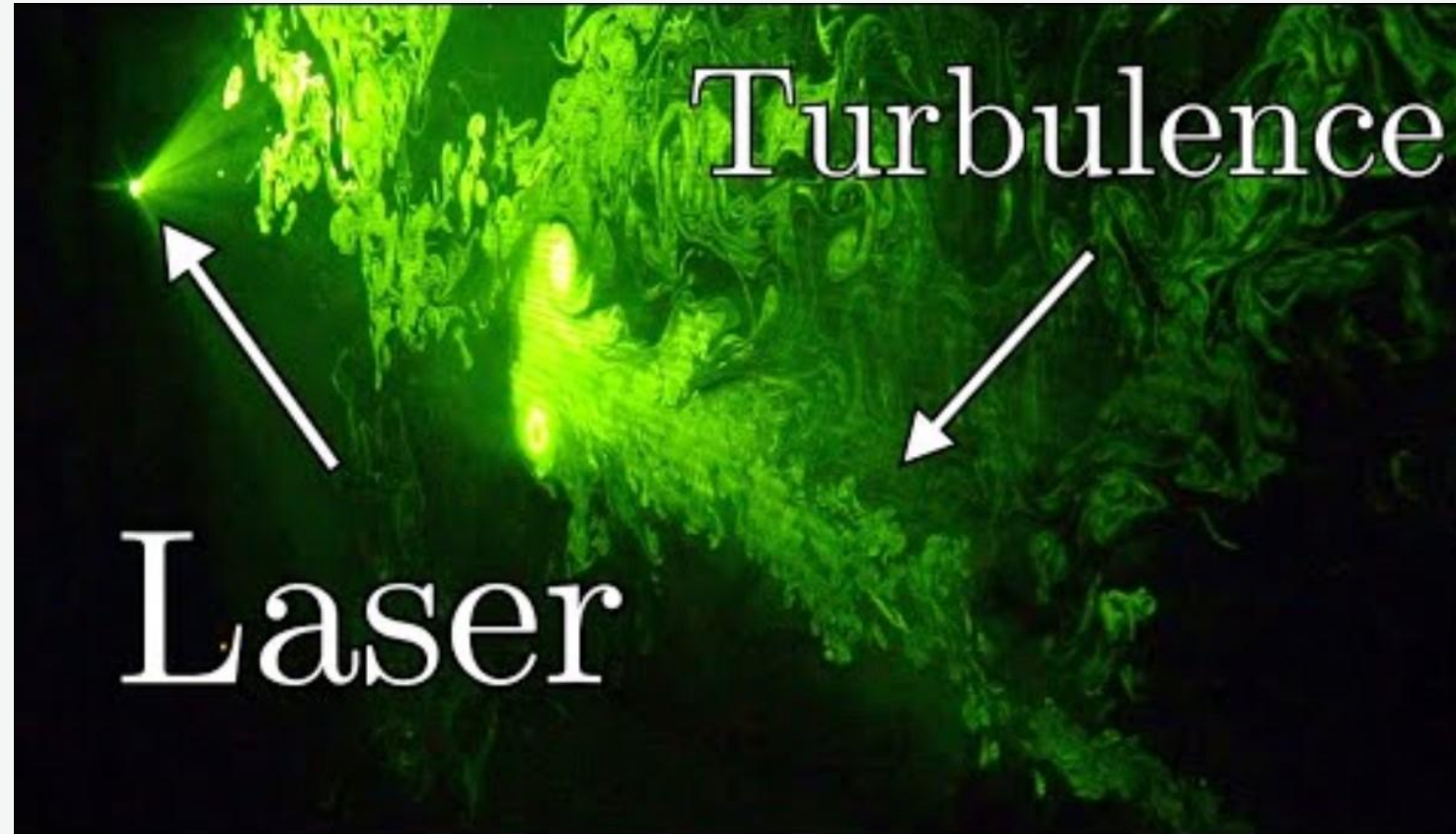
- Write an application and a library in C++
- Write a plain bullet point plan for solving a **PDE**
- Write the solver using **core OpenFOAM libraries**





OpenFOAM; what is that?!

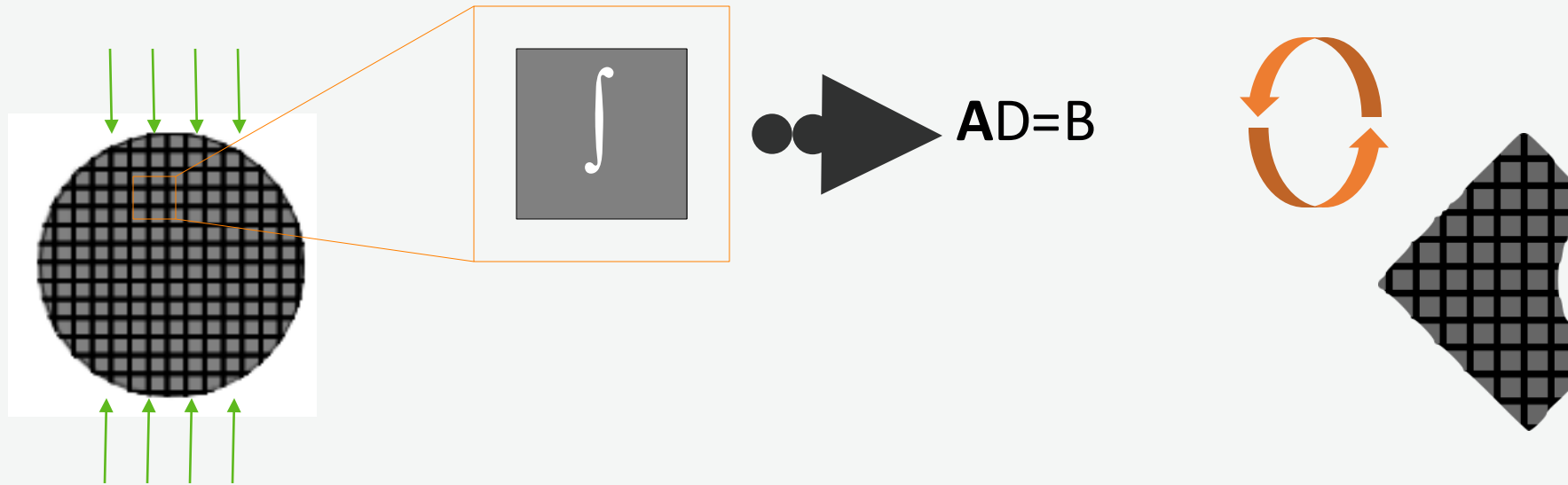
What is a **PDE**?



3Blue1Brown, Youtube



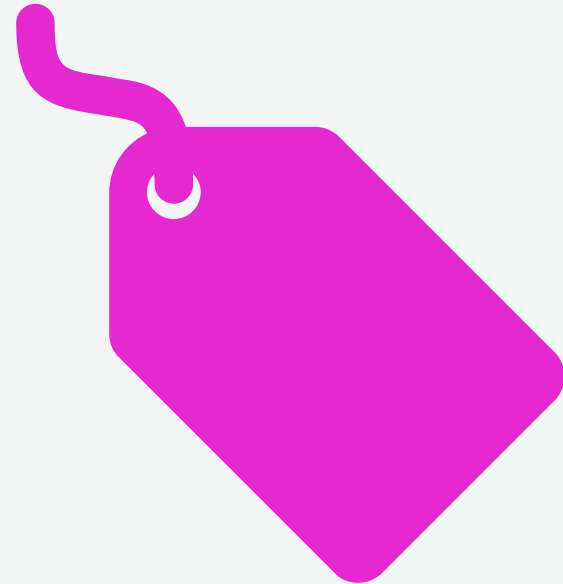
- A software to solve PDEs using Finite Volume Method



$$\rho \frac{d^2 u}{dt^2} = \nabla \cdot T + \rho f$$

Advantages

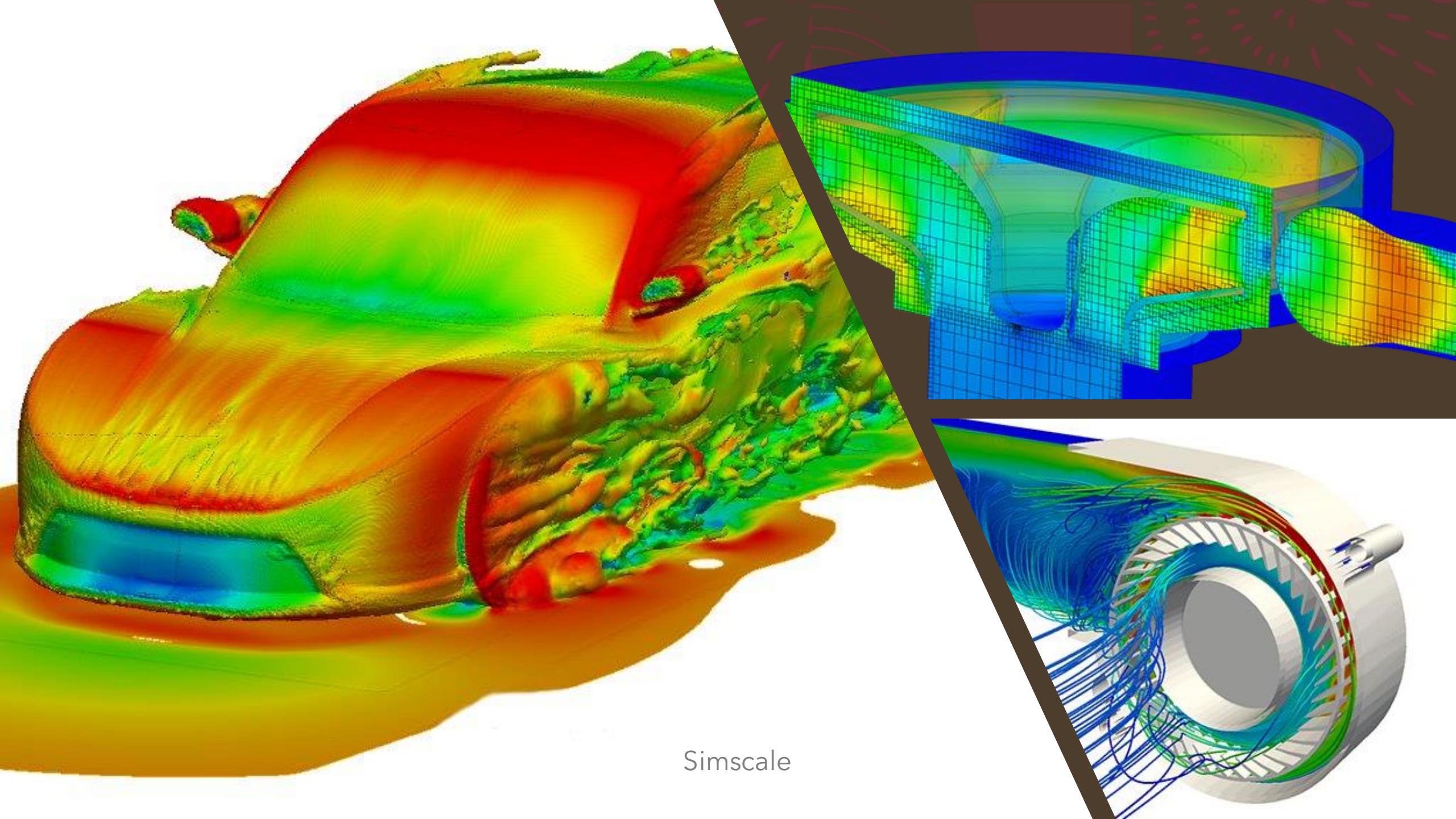
- Free price tag
- Open-source





Is It *Industry*-Standard?





Simscale

Any Downsides?



Any Downsides?



REVENUE STREAM



Any Downsides?



REVENUE STREAM



DOCUMENTATION



Any Downsides?



REVENUE STREAM

?



DOCUMENTATION



Linux

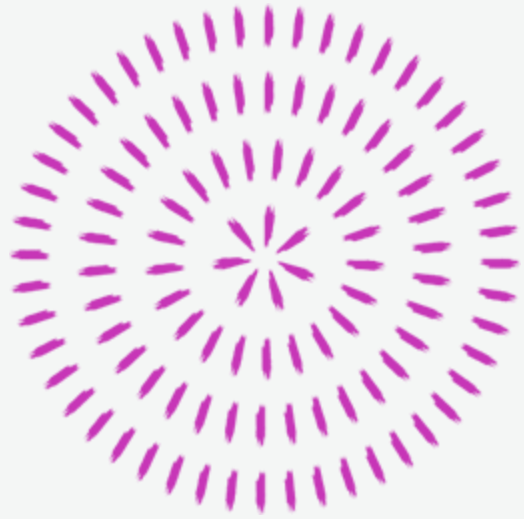
- Multiple distros
- Each distro releases new versions



Terminal

- Nautilus (file explorer)
- Terminal (command line)





A Quick Review of C++

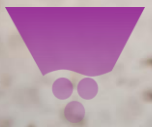




```
main()
```

Juicer :

lemonde

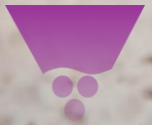




```
main()
```

Juicer :

```
lemonde juicer(lemons)
```





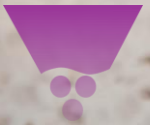
```
main()
```

Juicer :

```
lemonde juicer(lemons)
```

```
{
```

```
    peel(lemons);
```





```
main()
```

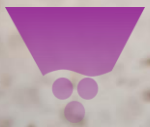
Juicer :

```
lemonde juicer(lemons)
```

```
{
```

```
    peel(lemons);
```

```
    cut(lemons);
```





```
main()
```

Juicer :

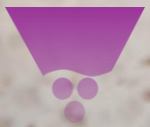
```
lemonde juicer(lemons)
```

```
{
```

```
    peel(lemons);
```

```
    cut(lemons);
```

```
    ...
```





main()

Juicer :

```
lemonade juicer(lemons)
{
    peel(lemons);
    cut(lemons);
    ...
    return (lemonade);
}
```



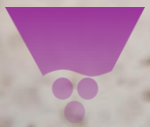

main()

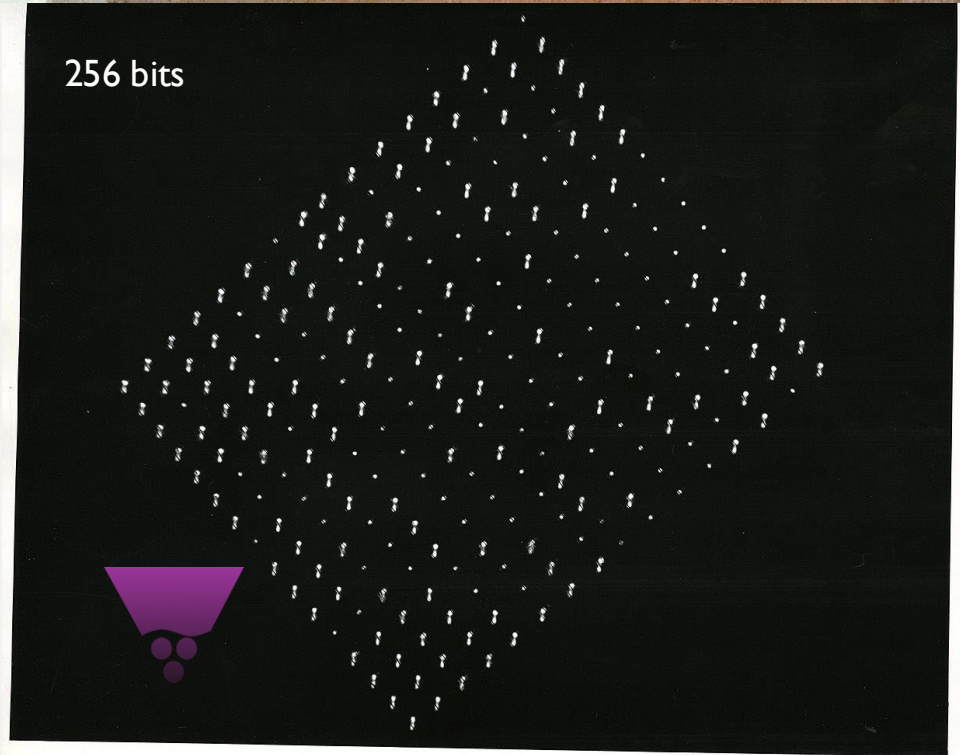
Juicer :

```
lemonde juicer(lemons)
{
    peel(lemons);
    cut(lemons);
    ...
    return (lemonade);
}
```

file.C :

```
int main() {}
```





main()

file.C :

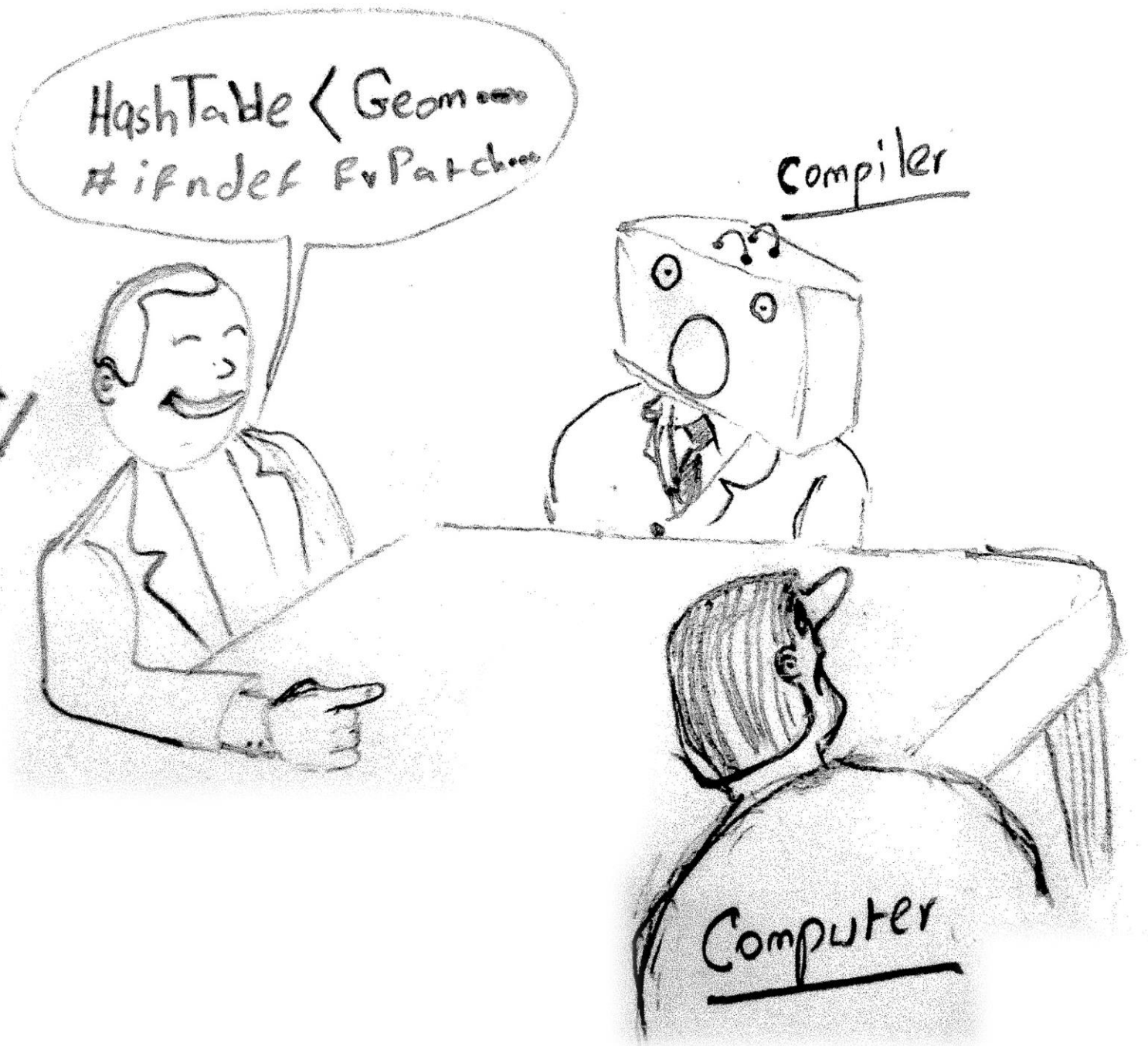
```
int main() {}
```

int
double

2 Bytes*
8 Bytes

* <https://cplusplus.com/doc/tutorial/variables>

Figures: https://en.wikipedia.org/wiki/Williams_tube



HashTable < Geom
#ifdef RvPatch...

Compiler

Developer

Computer

COMPILING: BASIC

simplest.C :

```
int main() {}
```

```
g++ -o file.o file.C
```





COMPILING: BASIC

simplest.C :

```
int main() {}
```

```
g++ -o file.o file.C
```

Object
code

Source
code



COMPILING: BASIC

simplest.C :

```
int main() {}
```

```
g++ -o file.o file.C
```

file.cc
file.cp
file.cxx
file.cpp
file.CPP
file.c++
File.C

(from man g++)



COMPILING: BASIC

simplest.C :

```
int main()
{
    std::cout << "Hello\n" << std::endl;
}
```




NAMESPACE



NAMESPACE

Sheer (in Farsi)





NAMESPACE

Sheer (in Farsi)





NAMESPACE

Sheer (in Farsi)



NAMESPACE

Sheer (in Farsi) ... ?



NAMESPACE

namespace food;

Sheer (in Farsi) ... ?

NAMESPACE

namespace food;

Sheer (in Farsi) ... ?



NAMESPACE

`namespace food;`

Sheer (in Farsi) ... ?

`namespace animal;`



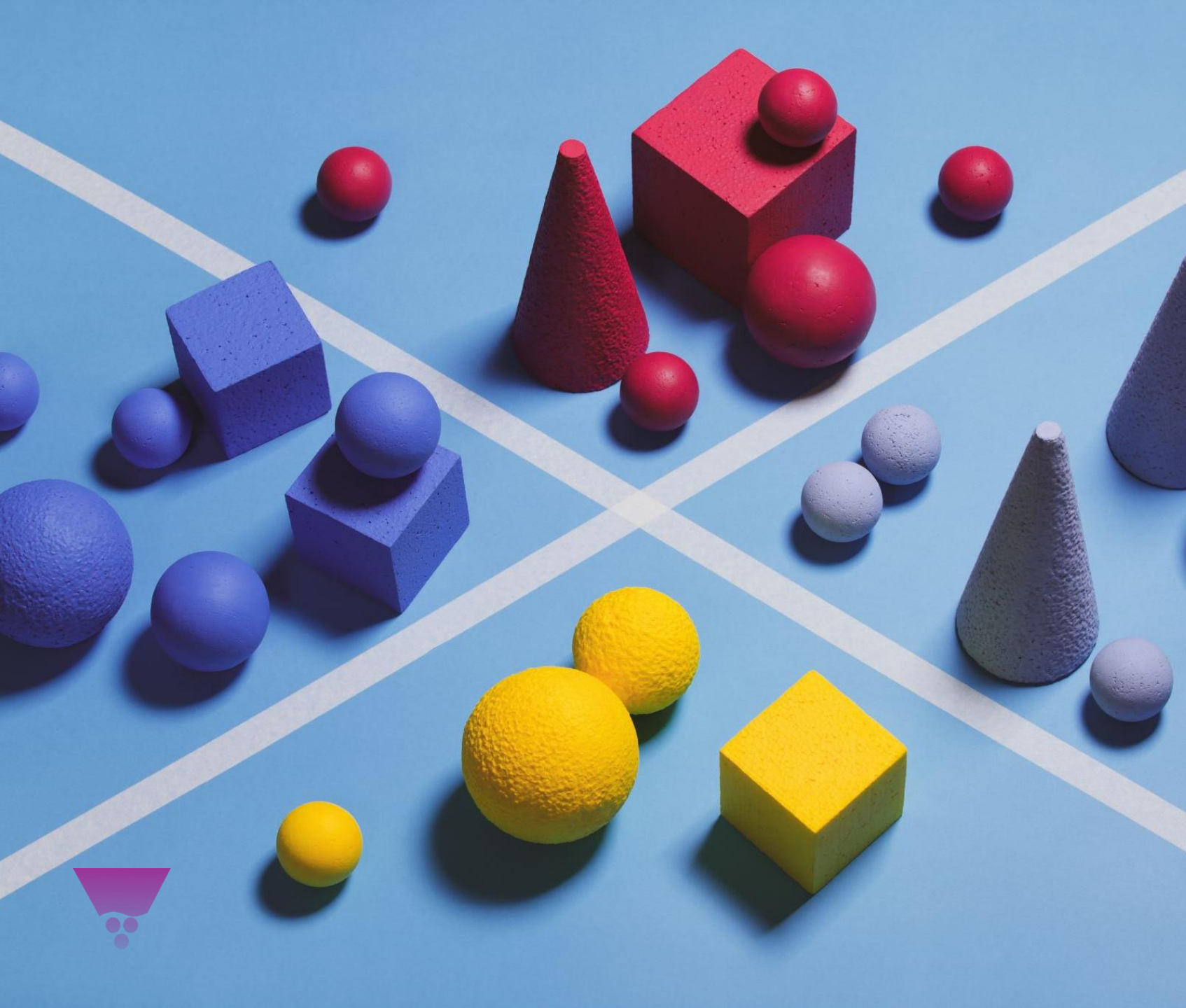
NAMESPACE

`namespace food;`

Sheer (in Farsi) ... ?

`namespace animal;`





OBJECTS



OOP

Objects



Brayan



Yanuar



Jawad

OOP

Class

student

Objects



Brayan



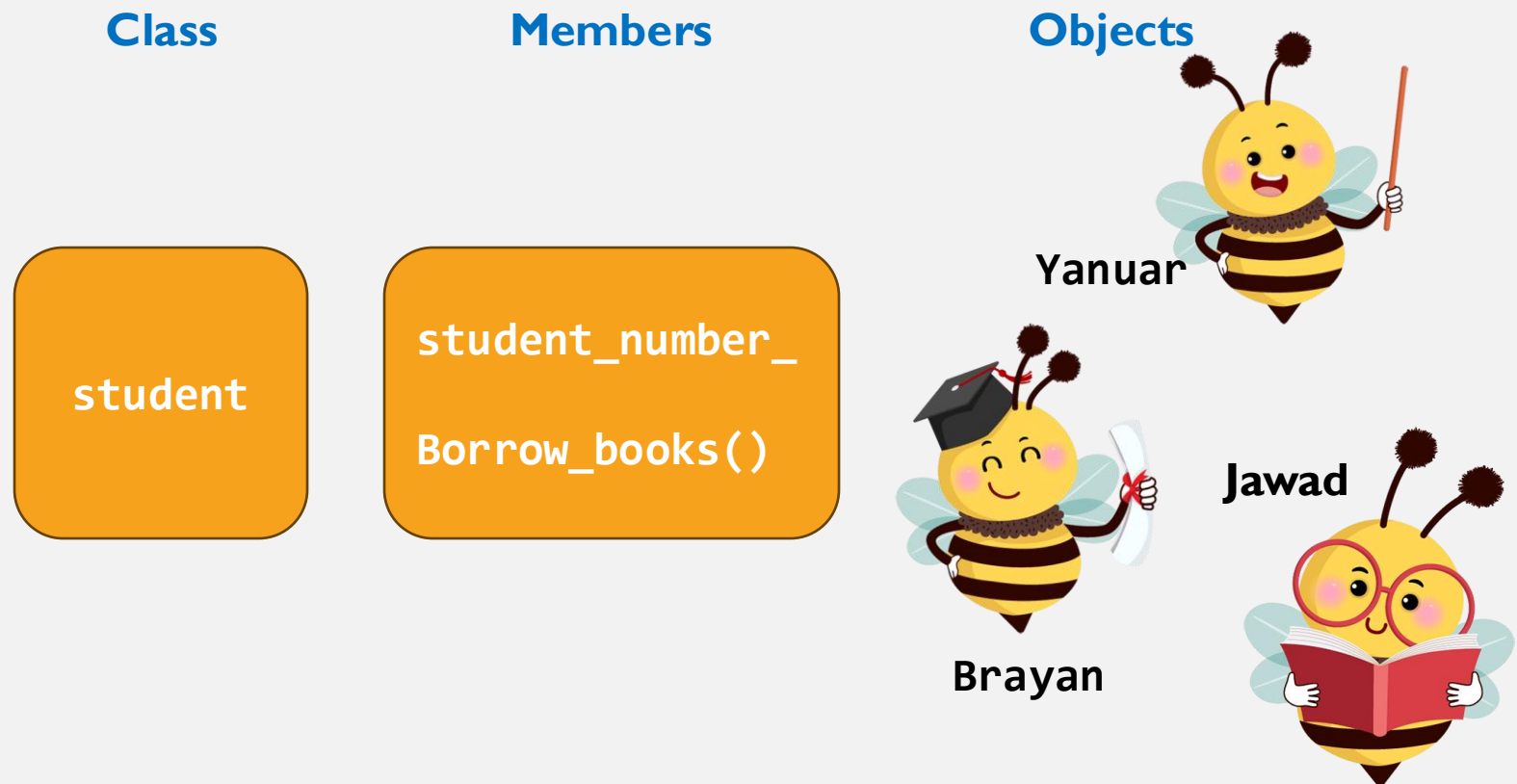
Yanuar



Jawad



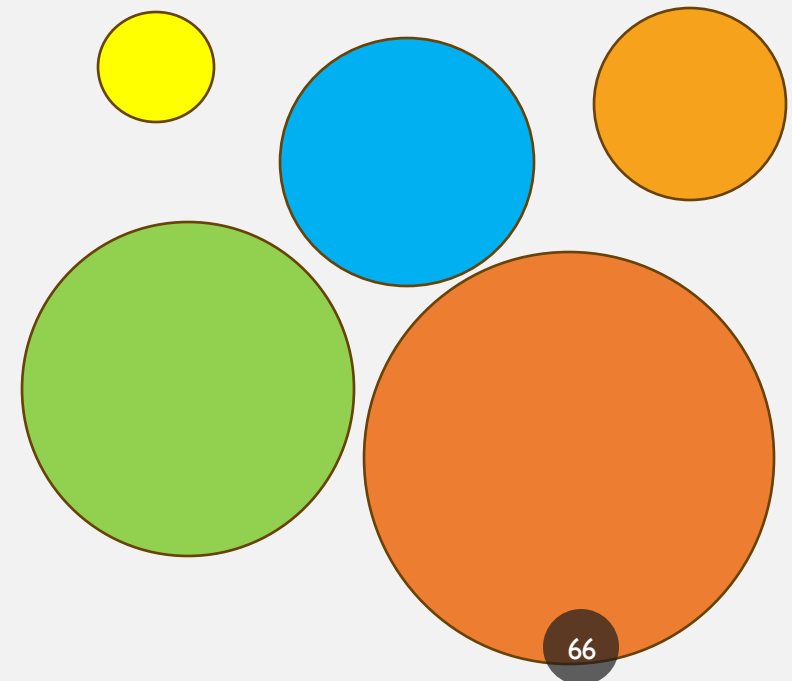
OBJECT-ORIENTED PROGRAMMING





OOP

Objects

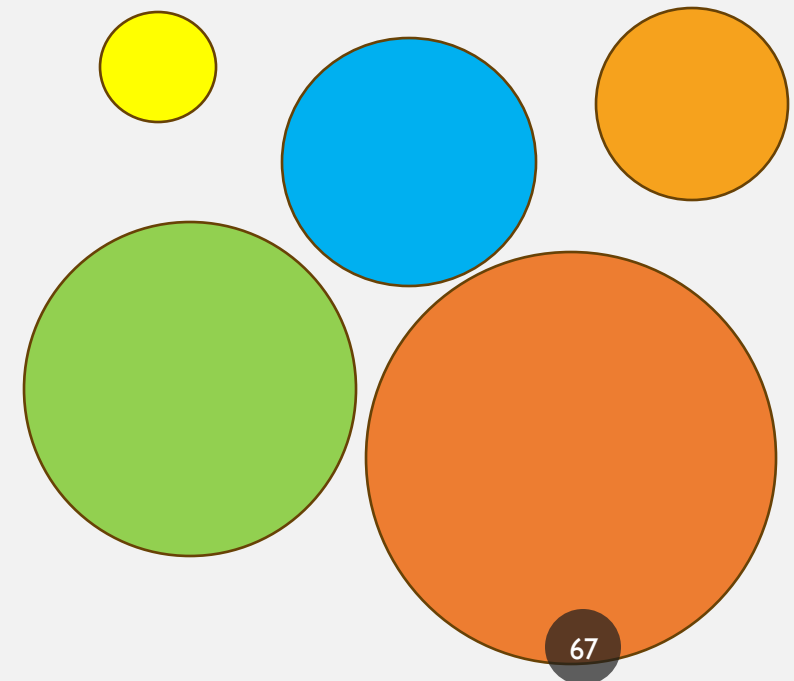


OOP

Class



Objects



OOP

Class

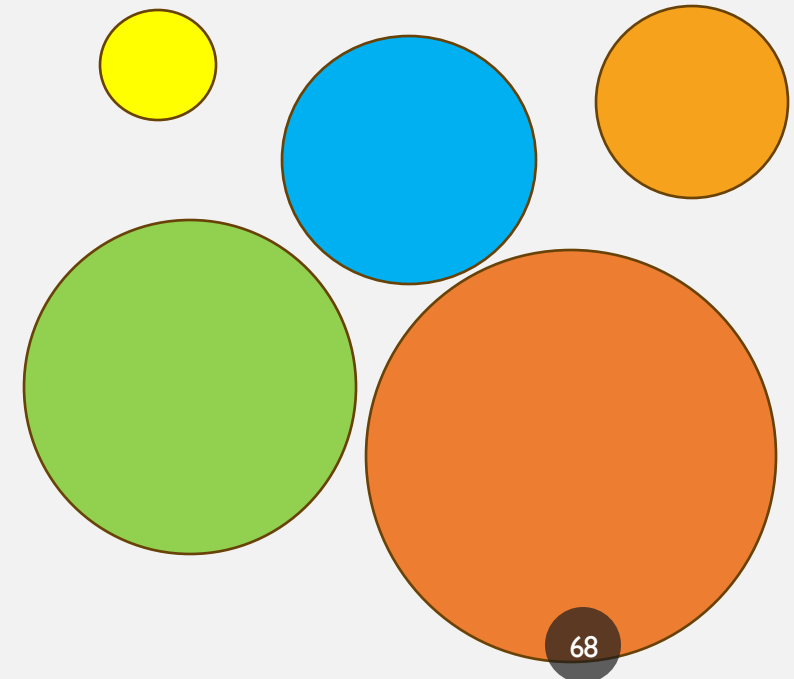


Attributes

Data
Diameter
Color

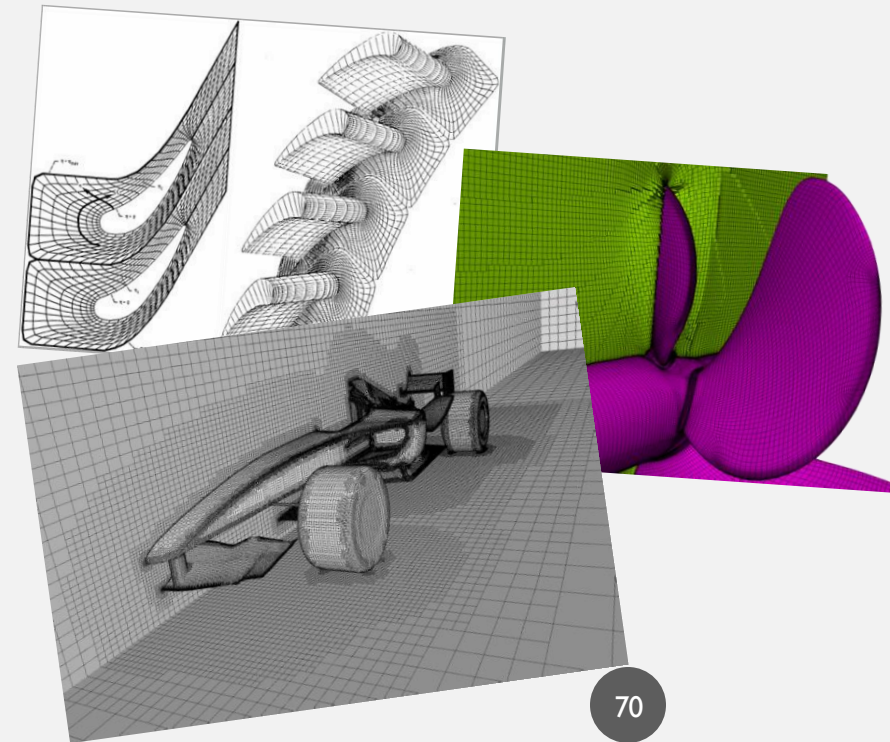
Methods
Area
Perimeter

Objects



OOP

Objects

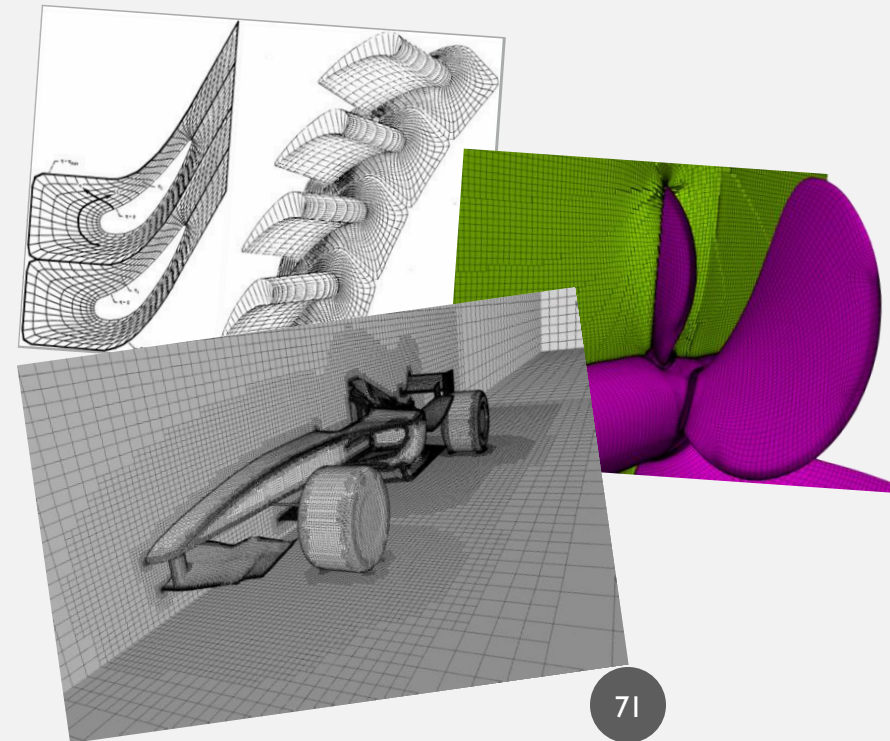


OOP

Class

fvMesh

Objects



OBJECT-ORIENTED PROGRAMMING

Class

fvMesh

Attributes

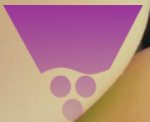
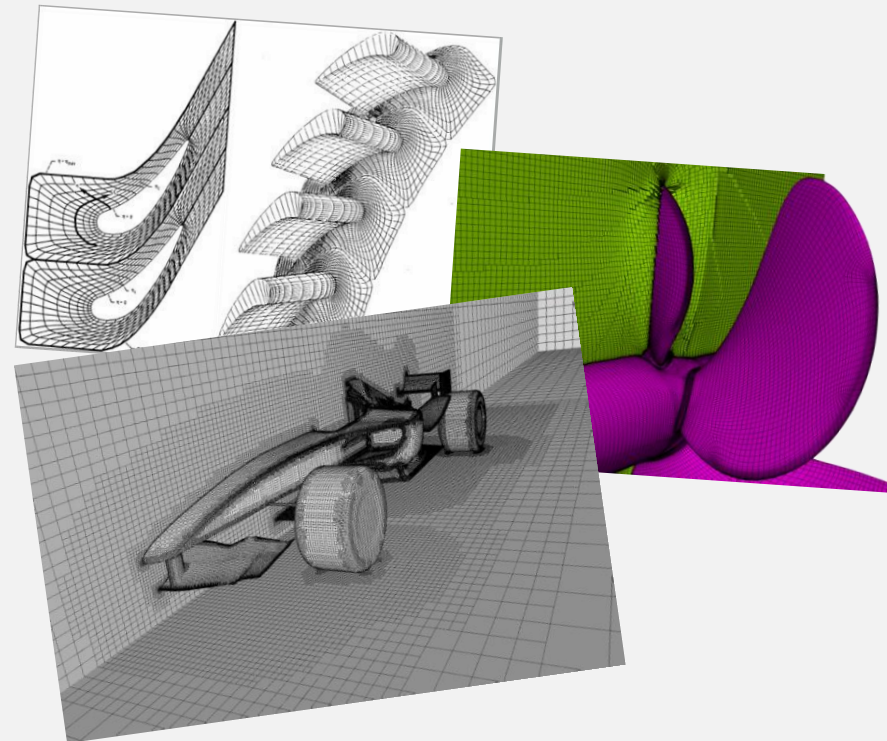
Data

points
faces
...

Methods

movePoints
addPatches

Objects





DRIVER

Calculate the area(A) of a circle with *diameter*(D)= d unit: *area.C*

```
int main()
{
    circle c1(1);
    std::cout << c1.area() << std::endl;

    return 0;
}
```




circle CLASS

Calculate the area(A) of a circle with *diameter*(D)= d unit: *area.C*

```
class circle
{
    // member data
    double d_;

    public:

    // member function(s)
    double area();

    // Constructor(s)
    circle(double);
};
```



circle CLASS

Calculate the area(A) of a circle with *diameter*(D)= d unit: *area.C*

```
circle::circle(double d):  
    d_(d)  
{  
  
    double circle::area()  
    {  
        return 3.14* d_ * d_ / 4;  
    }  
}
```



circle CLASS

Calculate the area(A) of a circle with *diameter*(D)= d unit:

Run





COMPILING:wmake

```
.
├── Make
│   ├── files
│   └── options
└── sayHello.C
```

Make/files:

sayHello.C

EXE=\$(FOAM_USER_APPBIN)/sayHello

QUIZZ

Create a class called **rectangle**;

Create a rectangle object;

Show its perimeter and area on terminal.



SOLUTION

We write a simple rectangle library

- See `src/rectangle`
- Compile: `cd src && wmake libso`

Now see the application below:

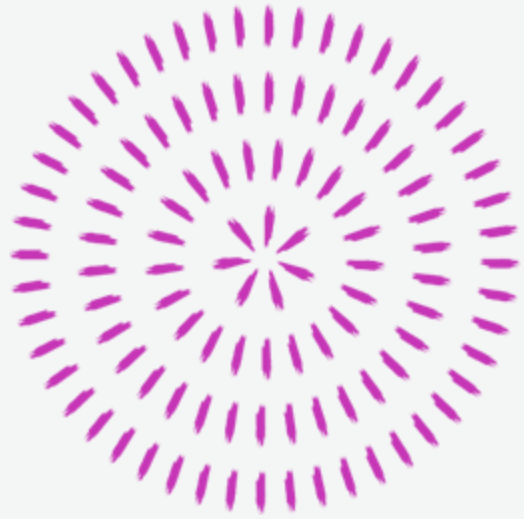
- One-time randomizer: `applications/calcRectangle/calcRectangle.C`

Compile :

- `cd applications/calcRectangle/ && wmake`

Test: Anywhere in your terminal run `calcRectangle`



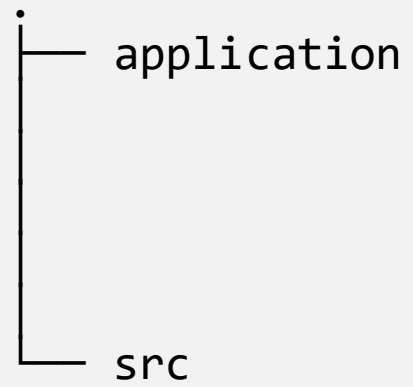


OpenFOAM Directory Tree





LIBRARY VS APPLICATION



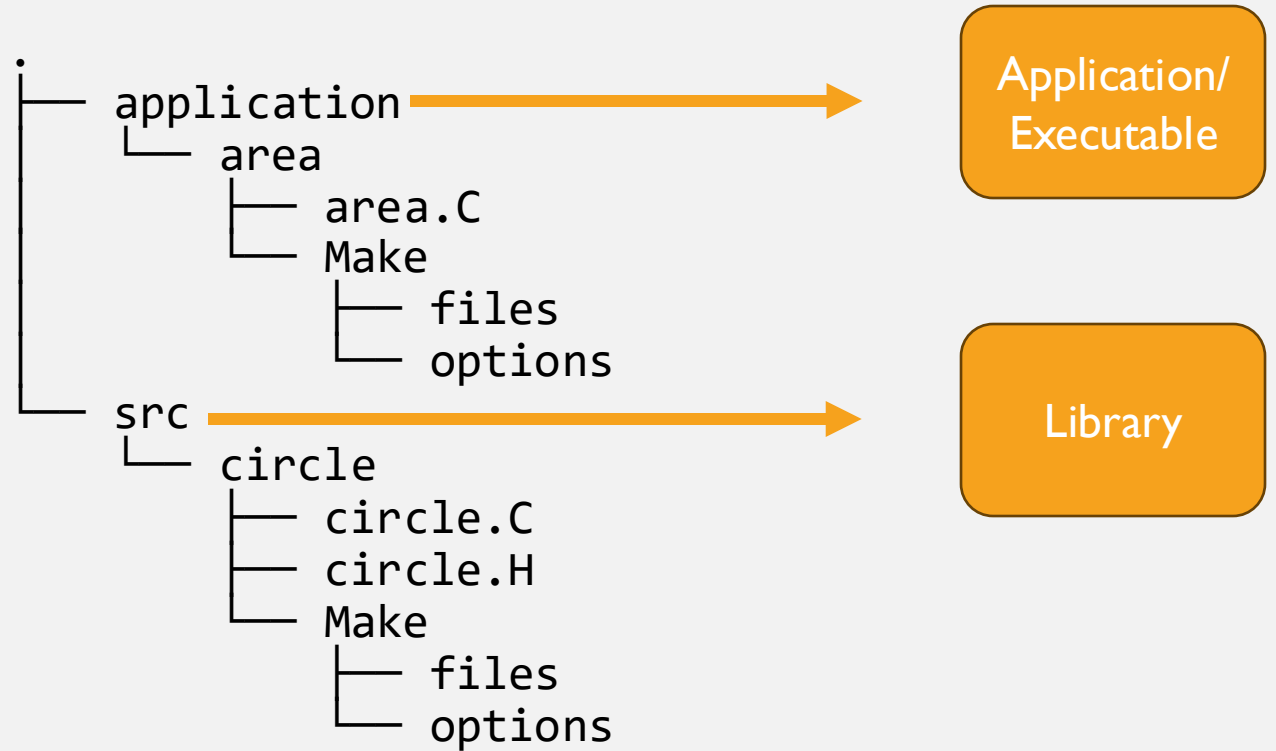


DIRECTORY TREE





LIBRARY VS APPLICATION





Thank You!

